

Model Dependence of PDG Comparison Values:

How Empirical Values Arise, Where Theory and Irrational Factors Enter, and What
This Means for FFGFT Residuals

Johann Pascher
ORCID 0009-0000-6518-4064

5 September 2026

Contents

Model Dependence of PDG Comparison Values	1
1 The question	2
2 How empirical values arise	2
The CODATA adjustment	2
Electron mass m_e	2
Muon-electron mass ratio m_μ/m_e	2
Fine-structure constant α	3
Tau mass m_τ — two methods	3
Fermi constant G_F and vacuum expectation value v	3
Gravitational constant G	4
Neutrino mass-squared differences	4
Overview	4
3 Irrational factors in the extraction formulae	4
4 The residual in $(m_\mu/m_e)^2$: exact computation	5
5 Error accumulation along the derivation chain	7
6 Consequences for the corpus	9
1 Affected derivation chains	11

Status markers

- [K]** *Confirmed* — derived from ξ , numerically verified.
- [B]** *Proved* — algebraically established.
- [S]** *Speculative* — open.
- [Q]** *Source* — corpus-external primary source, verified here.

Abstract

FFGFT predictions are compared against PDG and CODATA values. This document examines how those comparison values actually arise: which raw observable underlies each, which theory (QED, Standard Model) enters the extraction, and where irrational numbers already appear in the extraction formulae themselves.

Main findings: (1) m_e and m_μ/m_e are the least model-dependent anchors, but m_μ/m_e is QED-dependent and derives from a 27-year-old muonium HFS measurement whose theory uncertainty CODATA underestimates by a factor of 5–10 (Eides 2026). (2) α carries an unresolved 5σ tension between the two best recoil measurements. (3) $v = 246$ GeV is never measured; it is the SM definition $v \equiv (\sqrt{2} G_F)^{-1/2}$. (4) G_F carries a theory correction $\Delta q = -0.44\%$ with a 4.7σ hadronic tension. (5) m_τ has two measurement methods differing by 0.18 MeV; FFGFT lies 0.4σ from the PDG average. (6) The Galois identity $(m_\mu/m_e)^2 = 43200$ and the PDG masses agree very well: the ratio deviates by $-38.99 \cdot \xi \approx 0.52\%$ linearly (resp.

1.03 % quadratically). No known error source accounts for this residual — it is an empirical discrepancy not explained by the stated comparison uncertainty; its attribution to the Layer-1→Layer-2 transition remains bounded by the open **[S]**. (7) Errors accumulate along the derivation chain; every prediction requires its full provenance path.

All 21 primary sources were verified on 5 September 2026.

1 The question

The corpus compares FFGFT predictions with PDG values and finds residuals at the sub-percent to percent level. Are PDG values theory-neutral, or do they themselves depend on the Standard Model — such that part of the residual arises on the extraction side rather than the FFGFT side?

The FFGFT foundational statement: only ratios free of irrational numbers are exact; SI absolute values are structurally approximate (Docs. 241, 085). If PDG values themselves contain irrational factors in their extraction, the same limitation applies to the comparison side.

2 How empirical values arise

No PDG or CODATA value is a raw measurement. Each passes through the same chain:

raw observable → theory formula → extracted parameter → CODATA adjustment → PDG value

The CODATA adjustment

CODATA publishes recommended values every four years, produced by a joint least-squares adjustment over roughly 300 input data linked by theory relations (QED, Rydberg formula, kinematics). The values are therefore *correlated*: a change in one input shifts all dependent values. Where theory uncertainties enter, they are partly treated as adjustment parameters (δ_{th} in the observational equations) rather than as errors — this is the Eides finding for muonium HFS **[Q]**.

Electron mass m_e

Raw observable: cyclotron frequency of a hydrogen-like ion ($^{12}\text{C}^{5+}$, $^{28}\text{Si}^{13+}$, $^{20}\text{Ne}^{9+}$) in a Penning trap, and the Larmor precession frequency of the bound electron. The ratio is a pure number.

Theory formula: bound g -factor from QED bound-state theory. From the frequency ratio and g_{bound} follows m_e/m_{ion} ; with the ion mass in u follows m_e in u; conversion to MeV via $\text{u} \rightarrow \text{kg} \rightarrow \text{MeV}$.

Theory dependence: QED for g_{bound} , small and well controlled. Current: Heiße et al. (2023) at 0.1 ppb **[Q]**.

Muon-electron mass ratio m_μ/m_e

Raw observable: ground-state hyperfine splitting of muonium (a bound μ^+e^- state) — a microwave resonance at 4.463 GHz — and Zeeman splitting in a magnetic field.

How m_μ/m_e is measured. The muon is not placed on a scale. From the measured HFS frequency $\Delta\nu$ and the QED formula for the Fermi frequency

$$\nu_F = \frac{16}{3} \alpha^2 c R_\infty \frac{m_e}{m_\mu} \left(\frac{m_r}{m_e} \right)^3 \quad (1)$$

(corrections: QED to $O(\alpha^3)$, recoil in m_e/m_μ , weak Z exchange, hadronic VP) m_μ/m_e follows. Two consequences:

- m_μ/m_e is *not theory-free*: the extracted value contains assumptions about α , higher-order QED loop corrections, and hadronic vacuum polarisation. Since $\nu_F \propto \alpha^2$, α enters implicitly — anchor and α are not independent.
- *CODATA overstates the real precision*: CODATA 2022 gives $\sigma_{\text{rel}}(m_\mu/m_e) = 2.2 \times 10^{-8}$. The actual uncertainty of the HFS extraction (experiment + QED theory from Eides 2026 + α tension) is $\approx 1.2 \times 10^{-7}$ — a factor of 5–6 larger. **[Q]**

Theory dependence: high. Sole precision measurement: LAMPF 1999 (Liu et al.). Eides (2026) shows that CODATA lists the measured value 4 463 302 776(51) Hz as the theoretical value; the actual theory uncertainty is 271–515 Hz **[Q]**.

Fine-structure constant α

Two routes that disagree at 5σ **[Q]**:

- **Atom recoil:** $\alpha^2 = 2R_\infty (h/m_{\text{atom}}) (m_{\text{atom}}/m_e)/c$. Rb (Morel et al. 2020): $\alpha^{-1} = 137.035\,999\,206(11)$. Cs (Parker et al. 2018): $137.035\,999\,046(27)$. Difference 1.6×10^{-7} , $> 5\sigma$.
- **Electron $g - 2$:** a_e at 10^{-13} (Fan et al. 2023) plus 5-loop QED (12 672 diagrams, Aoyama et al.). Gives $\alpha^{-1} = 137.035\,999\,166(15)$ — between Rb and Cs.

The CODATA value $137.035\,999\,177(21)$ is an average covering none of the measurements. FFGFT: $\alpha^{-1} = 3700/27 = 137.037\,037\dots$ (Doc. 338, R101), deviation 7.6 ppm — $100\times$ larger than the Rb/Cs tension.

Tau mass m_τ — two methods

Method	Experiment	Value (MeV)	Theory assumption
Threshold scan	BESIII 2014	$1776.91 \pm 0.12^{+0.10}_{-0.13}$	QED cross-section
Pseudomass	Belle II 2023	$1777.09 \pm 0.08 \pm 0.11$	$m_{\nu_\tau} = 0$
PDG average 2024	—	1776.93 ± 0.09	—

Threshold scan: $\sigma(e^+e^- \rightarrow \tau^+\tau^-)$ versus beam energy near $2m_\tau \approx 3.55$ GeV; m_τ as fit parameter of the QED threshold curve. **Pseudomass:** at 10.6 GeV, the kinematic edge of the invariant mass in $\tau \rightarrow 3\pi\nu_\tau$ assumes $m_{\nu_\tau} = 0$.

For FFGFT: $m_\tau = 1776.97$ MeV (pred) lies 0.04 MeV $= 0.4\sigma$ from the PDG average. The methods differ by 0.18 MeV. The falsification test is undecided; Belle II with > 1 ab $^{-1}$ should reach $\sigma < 0.05$ MeV.

Fermi constant G_F and vacuum expectation value v

Raw observable for G_F : muon decay time distribution (MuLan, 10^{12} decays), $\tau_\mu = 2.196\,980\,3(22)$ μs **[Q]**.

Theory formula (Eberhart et al. 2026 **[Q]**):

$$\tau_\mu^{-1} = \frac{G_F^2 m_\mu^5}{192\pi^3} (1 + \Delta q), \quad \Delta q = (-4\,384\,678 \pm 34) \times 10^{-9} \approx -0.44\%. \quad (2)$$

Δq contains QED to $O(\alpha^3)$, hadronic vacuum polarisation, electron-mass effects. Purely electroweak corrections are absorbed into G_F by definition (Sirlin 1980). The hadronic component carries a 4.7σ tension (CMD-3 vs. older data).

Raw observable for ν : none. $\nu \equiv (\sqrt{2} G_F)^{-1/2} = 246.22$ GeV follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$. **ν exists only within the Standard Model.** Its numerical value is the muon lifetime in SM clothing.

For FFGFT: Doc. 006/R104 gives $\nu = 248.3$ GeV (without K_{frak}) or 245.6 GeV (with $K_{\text{frak}} = 74/75$). Comparison with 246.22 GeV is a comparison of *FFGFT structure against SM definition*, not against an observable. Only τ_μ and m_μ are measured.

Gravitational constant G

Torsion balance (Li et al. 2018, 11.6 ppm) or atom interferometry (Rosi et al. 2014, 150 ppm). Theory: Newton. The only model-free value, but the least precise: CODATA $6.67430(15) \times 10^{-11}$, 22 ppm **[Q]**.

Neutrino mass-squared differences

Event rates versus L/E , three-flavour fit (NuFIT 6.0 **[Q]**). Δm_{31}^2 depends on whether SK atmospheric data are included: ratio 33.7 (with) vs. 33.9 (without). FFGFT $360/11 = 32.73$; measurement uncertainty 1–3 % $\approx$ FFGFT residual. Not discriminating.

Overview

Quantity	Raw observable	Theory in extraction	Convention
m_e	frequency ratio	QED (g_{bound})	—
m_μ/m_e	HFS frequency	QED, weak, hadr.	—
α (Rb/Cs)	interferometer phase	R_∞ (QED)	—
α ($g-2$)	frequency ratio	5-loop QED	—
m_τ	rates vs. E / edge	QED threshold / $m_{\nu_\tau} = 0$	beam calibration
G_F	decay times	Fermi + QED + hadr.	Sirlin absorption
ν	—	—	$\nu \equiv (\sqrt{2} G_F)^{-1/2}$
G	angle / acceleration	Newton	—
Δm^2	rates vs. L/E	3-flavour	with/without SK-atm

No value is theory-free. For FFGFT this means: comparison with PDG values is comparison with “QED + convention”, not with raw observations.

3 Irrational factors in the extraction formulae

The FFGFT foundational statement — ratios of rationals are exact, SI absolute values carry the approximation of the projection $S_m^1 \rightarrow \mathbb{R}_t$ (Docs. 306, 333) — applies to the SM extraction formulae themselves:

Quantity	Formula	Irrational factor
G_F	$\tau_\mu^{-1} = G_F^2 m_\mu^5 / (192\pi^3)$	π^3
ν	$\nu = (\sqrt{2} G_F)^{-1/2}$	$\sqrt{2}$
α (recoil)	$\alpha^2 = 2R_\infty h / (mc)$	$R_\infty \propto \alpha^2$: implicitly circular
HFS	$\nu_F \propto \alpha^2$	α^2
Δq	expansion in α/π	π in every term

" G_F measured" means: τ_μ measured, divided by $192\pi^3$, square root taken. " ν measured" additionally means: divided by $\sqrt{2}$, square root again. The irrational factors come from the SM formulation (phase-space integral, Higgs normalisation), not from measurement. The only convention-free comparison would be at the level of $\tau_\mu = 2.196\,980\,3\,\mu\text{s}$ — which FFGFT would have to predict directly **[S]**.

4 The residual in $(m_\mu/m_e)^2$: exact computation

FFGFT bare (Doc. 338): $(m_\mu/m_e)^2 = 43200$ exact, integer, $= |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$. With CODATA 2022 in exact rational arithmetic:

$$(m_\mu/m_e)_{\text{PDG}}^2 = 42753.12 \quad (3)$$

$$\text{Residual} = 446.88 \quad (1.045\%). \quad (4)$$

Possible error sources, computed:

- CODATA uncertainty $(m_\mu/m_e)^2$: ± 0.002 — residual remains $240\,000\times$ larger.
- Eides correction of HFS theory uncertainty (515 instead of 51 Hz): ± 0.010 — residual remains $45\,000\times$ larger.
- α choice in HFS anchor (FFGFT 3700/27 instead of CODATA): shifts $(m_\mu/m_e)^2$ by -1.3 — residual remains $340\times$ larger.
- To reach 43200, m_μ would need to be 551 keV higher (measurement error 2.3 eV), or m_e 2.65 keV lower (measurement error 0.15 meV), or the HFS 23 MHz higher (measurement error 51 Hz).

Conclusion [B]: No measurement error explains the residual. The discrepancy is an empirical one not accounted for by the stated uncertainty of the comparison value. Its physical attribution is left to the open **[S]**. The difference axis $\varepsilon_\mu - \varepsilon_e \approx -39 \cdot \xi$ measures the ratio residual, the product residual $\approx -1.2 \cdot \xi$ the sum axis. Both are empirically computed **[K]**.

Two orthogonal axes: difference and sum of logarithms

The PDG masses m_e and m_μ produce two independent logarithmic projections when compared with the Galois identities:

$$\underbrace{\ln \frac{m_\mu/m_e}{\sqrt{43200}}}_{\text{difference axis}} = \varepsilon_\mu - \varepsilon_e \approx -38.99 \cdot \xi \quad \underbrace{\ln \frac{m_e \cdot m_\mu}{54 \text{ MeV}^2}}_{\text{sum axis}} = \varepsilon_e + \varepsilon_\mu \approx -1.21 \cdot \xi$$

Difference axis: linear and square are one single finding. The linear ratio m_μ/m_e vs. $\sqrt{43200}$ (-0.519%) and the square $(m_\mu/m_e)^2$ vs. 43200 (-1.034%) lie on *exactly the same axis* $\varepsilon_\mu - \varepsilon_e$. Logarithmically it holds exactly:

$$\ln \frac{(m_\mu/m_e)^2}{43200} = 2 \cdot \ln \frac{m_\mu/m_e}{\sqrt{43200}}.$$

The square doubles the percentage but measures no new physics. It is *one* finding, two ways of writing it. The value $-38.99 \cdot \xi$ is empirical; it lies 8σ (Eides) from $39 \cdot \xi$ and is not a Galois integer **[S]**.

Sum axis: the product is orthogonal. $\varepsilon_e \approx +18.9 \cdot \xi$ and $\varepsilon_\mu \approx -20.1 \cdot \xi$ are nearly equal in magnitude and opposite in sign. In the sum they nearly cancel; what remains is $-1.21 \cdot \xi$ — the common scale deviation of both masses. This is at order ξ , consistent with the SI

projection $S_m^1 \rightarrow \mathbb{R}_t$ (Docs. 085, 306, 333) **[K]**. The sum $(-1.21 \cdot \xi)$ does not follow from the difference $(-38.99 \cdot \xi)$ — they are two independent physical pieces of information.

Combination	Logarithm	Residual	Status
$\ln(m_\mu/m_e/\sqrt{43200})$	$\varepsilon_\mu - \varepsilon_e$	$-38.99 \cdot \xi$	ratio residual [K] ; cause open [S]
$\ln((m_\mu/m_e)^2/43200)$	$2(\varepsilon_\mu - \varepsilon_e)$	$-77.99 \cdot \xi$	exactly = 2× above
$\ln(m_e \cdot m_\mu/54 \text{ MeV}^2)$	$\varepsilon_e + \varepsilon_\mu$	$-1.21 \cdot \xi$	follows from anchor + ratio residual [K]

The apparently different percentages -0.016% , -0.52% , and -1.03% arise from two numbers: the generation tension $(-39 \cdot \xi)$ and the SI projection $(-1.2 \cdot \xi)$. Their large ratio is not an amplification but a cancellation in the sum (verification script: `pruef_351_generationsdifferenz.py`).

The residual of the product comparison lies at order ξ itself:

Quantity	Residual in units of ξ	Interpretation
$m_e \cdot m_\mu$ vs. 54	$-1.21 \cdot \xi$	order ξ : SI projection [K]
$(m_\mu/m_e)^2$ vs. 43200	$-77.6 \cdot \xi$	ratio residual, order 100ξ [K]

The SI projection $S_m^1 \rightarrow \mathbb{R}_t$ (Docs. 085, 306, 333) structurally produces errors of order ξ . The product residual $-0.016\% \approx 1.2 \cdot \xi$ lies precisely in this range and is therefore structurally explainable as an SI transition approximation **[K]**. The quadratic residual $-1.034\% \approx 78 \cdot \xi$ lies in higher order — ratio residual, empirically computed **[K]**.

Result: The Galois identity $m_e \cdot m_\mu = 54 \text{ MeV}^2$ is the only currently known identity whose residual lies at order ξ — the structurally expected SI projection error. The residual is therefore not unexplained; it corresponds to the approximation of the SI transition.

Constitutional description (after Stefaan Vossen, 5 Sept. 2026)

The following four-layer formulation summarises the epistemic status of the FFGFT comparisons:

Layer	Status
Layer-1	Exact within rational arithmetic: $(m_\mu/m_e)^2 = 43200$ and $ GF(3)^* \cdot GF(27) = 54$ are integer Galois identities without approximation [B] .
Layer-2	One empirical dimensional anchor: $m_\mu = 105.66 \text{ MeV}$ (PDG). All absolute predictions follow from it [K] .
Residual	Empirical discrepancy with a structural explanation of approximately the correct order: $(m_\mu/m_e)^2$ deviates by $\approx 78 \cdot \xi$ from the PDG value (ratio residual, order 100ξ); $m_e \cdot m_\mu$ deviates by $\approx 1.2 \cdot \xi$ (follows from anchor + ratio residual) [K] .
Quantitative bridge	The forward derivation of the exact residual from the $S_m^1 \rightarrow \mathbb{R}_t$ non-isomorphism is not in the corpus. Open [S] .

The open [S] is not a gap in the pejorative sense but a precisely constituted limit: the theory states explicitly where its formal construction stops.

Methodological note (after Stefaan Vossen, 5 Sept. 2026)

"Constitution does not tell the Galois identity what it is. It tells us what a particular scientific use of that identity is presently entitled to mean."

Two distinctions to be preserved in the record:

1. **54 as a finite-field integer $\neq$ 54 MeV² as a dimensional representation.** The identity $|GF(3)^*| \cdot |GF(27)| = 2 \cdot 27 = 54$ belongs to the dimensionless Layer-1 algebra **[B]**. $m_e \cdot m_\mu = 54 \text{ MeV}^2$ is a dimensional representation produced by the mapping into SI units (Layer-2). Both are distinguishable scientific objects; the unit MeV² belongs to the dimensional mapping, not to the dimensionless algebra alone.
2. **Numerical correspondence vs. physical attribution.** The residual $m_e \cdot m_\mu \approx 54 \text{ MeV}^2$ with deviation $\approx 1.2 \cdot \xi$ is an established numerical correspondence **[K]**. The physical attribution of that residual to the SI projection $S_m^1 \rightarrow \mathbb{R}_t$ (Docs. 085, 306, 333) remains bounded by the still-open quantitative bridge **[S]**. The correspondence stands; the attribution is a proposal, not a derivation.

Methodological remark: relation vs. prediction

The three percentages (-0.519% , -1.034% , -0.016%) are *not tolerance values*. Tolerance arises only when a **calculated** mass deviates from a **measured** one — which requires an independent FFGFT prediction for m_e or m_μ individually.

The Galois identities 43200 and 54 MeV² are **relations** between both masses, not individual predictions. As long as no independent FFGFT value for m_μ (or m_e) exists, none of the three percentages has a defined limit.

Moreover: every direct mass carries the fractal correction K_{frak} and the SI projection $S_m^1 \rightarrow \mathbb{R}_t$. In the ratio, K_{frak} cancels; the SI corrections *subtract* (difference axis, $\approx -39 \cdot \xi$). In the product, K^2 remains; the SI corrections *add* (sum axis, $\approx -1.2 \cdot \xi$). The different percentages arise from this — not from measurement errors.

No independent single prediction for m_μ without an input anchor exists in the corpus. With m_e as anchor (SETZUNG) there are two routes:

- **Square route:** $m_\mu = m_e \cdot \sqrt{43200} = m_e \cdot 120\sqrt{3}$ — irrational, deviation $+0.52\%$. Not a clean Layer-1 value.
- **Product route:** $m_\mu = 54 \text{ MeV}^2 / m_e$ — rational, deviation $+0.016\% \approx 1.2 \cdot \xi$. The only rational tolerance value for m_μ at anchor m_e .

But: $m_{e,\text{PDG}}$ already carries K_{frak} and the SI projection. Hence $m_\mu^{\text{pred}} = 54 / m_{e,\text{PDG}} \approx m_{\mu,\text{bare}}(1 - \varepsilon_e)$ and $m_{\mu,\text{PDG}} \approx m_{\mu,\text{bare}}(1 + \varepsilon_\mu)$, so:

$$\frac{m_\mu^{\text{pred}}}{m_{\mu,\text{PDG}}} - 1 \approx -(\varepsilon_e + \varepsilon_\mu) \approx -1.2 \cdot \xi.$$

This is the sum axis — the same relation as the product residual. The fractal correction does not cancel but is contained in ε_e and ε_μ and adds in the product. A true Layer-1 tolerance value for m_μ without any correction does not exist, because every direct mass carries K_{frak} and the SI projection.

5 Error accumulation along the derivation chain

Every derivation step introduces a residual. What matters is which combination is formed, because that determines which corrections remain visible and which cancel (Section 4).

First step $m_e \rightarrow m_\mu$ **rational route.** With m_e as anchor (SETZUNG) and the Galois identity $m_e \cdot m_\mu = 54 \text{ MeV}^2$, one obtains rationally:

$$m_\mu^{\text{pred}} = \frac{54 \text{ MeV}^2}{m_{e,\text{PDG}}} = 105.675 \text{ MeV}, \quad \text{residual: } +0.016 \% \approx +1.2 \cdot \xi \text{ [K]}.$$

This is the sum axis $(\varepsilon_e + \varepsilon_\mu)$; both masses carry K_{frak} and SI projection, which add in the sum. The irrational route $m_\mu = \sqrt{43200} m_e = 120\sqrt{3} m_e$ (+0.52 %) is not a clean Galois value (bookkeeping rule above).

Accumulation. For m_τ (next step via $m_\tau/m_\mu = 16.992$ [K], Doc. 317), for ν (SM definition, not measured, R112) and α (5σ tension Rb/Cs, R113) the uncertainties of the extraction paths accumulate. To determine where a residual originates, the convergence of empirical values to algebraic ones would need to be established — it is not.

Bookkeeping: For every reported prediction state: (1) the anchor, (2) the number of derivation steps, (3) corrections applied, (4) the comparison value with extraction method. "FFGFT predicts m_μ to 0.5 %" is incomplete; complete: "From m_e (PDG), in one step via $m_e \cdot m_\mu = 54 \text{ MeV}^2$, m_μ follows with +0.016 % relative to the PDG value."

σ balance of the prediction comparison. The prediction $m_\mu^{\text{pred}} = 54 \text{ MeV}^2 / m_{e,\text{PDG}}$ carries only the uncertainty of m_e :

$$\sigma(m_\mu^{\text{pred}}) = \frac{54 \text{ MeV}^2}{m_e^2} \cdot \sigma(m_e) = 3.1 \times 10^{-8} \text{ MeV} \quad (\sigma_{\text{rel}} = 2.9 \times 10^{-10}).$$

The comparison uncertainty combines in quadrature:

$$\sigma_{\text{comp}}^2 = \sigma(m_\mu^{\text{pred}})^2 + \sigma(m_{\mu,\text{PDG}})^2 \approx \sigma(m_{\mu,\text{PDG}})^2.$$

The contribution of $\sigma(m_e)$ is 0.25 % of σ_{comp} — negligible. The entire measurement uncertainty resides in $m_{\mu,\text{PDG}}$ (HFS extraction, Eides 2026). The deviation +0.016 % lies 7386σ (CODATA) or 1386σ (Eides) outside — the same as the product residual, now expressed as a σ distance.

Classification of the product residual. The product residual $-1.2 \cdot \xi$ is at order ξ . The MeV scale is exact since 2019 (h, c, e fixed); the measurement uncertainties $\sigma_{\text{rel}}(m_e) \approx 0$, $\sigma_{\text{rel}}(m_\mu) \approx 0.0002 \cdot \xi$ do not explain the residual. It follows from the Layer-1 $\leftrightarrow$ Layer-2 gap (near-complete cancellation of ε_e and ε_μ in the sum axis). The σ -range within which this residual is expected to lie has not been derived [S].

Algebraic dependency: only one open number

With anchor m_e (SETZUNG) and the Galois ratio $(m_\mu/m_e)^2 = 43200$ (Layer-1), it follows necessarily:

$$m_\mu = m_e \cdot \sqrt{43200}, \quad m_e \cdot m_\mu = m_e^2 \cdot \sqrt{43200} = 54.273 \text{ MeV}^2.$$

The product residual from these two inputs alone is $+37.9 \cdot \xi$ — no free parameter, no [S].

The PDG product residual is $-1.2 \cdot \xi$. The difference is $39.1 \cdot \xi$ — exactly the ratio residual. This is algebraically necessary:

$$\underbrace{(\varepsilon_e + \varepsilon_\mu)}_{\text{product residual } -1.2 \cdot \xi} = \underbrace{2\varepsilon_e}_{\text{from anchor } +37.9 \cdot \xi} + \underbrace{(\varepsilon_\mu - \varepsilon_e)}_{\text{ratio residual } -39.1 \cdot \xi}.$$

The product residual is not a third independent number — it is the sum of anchor contribution and ratio residual. There is only **one** open number: the ratio residual $-38.99 \cdot \xi$. The sole [S] sits there.

6 Consequences for the corpus

No numerical value in the corpus changes. What changes:

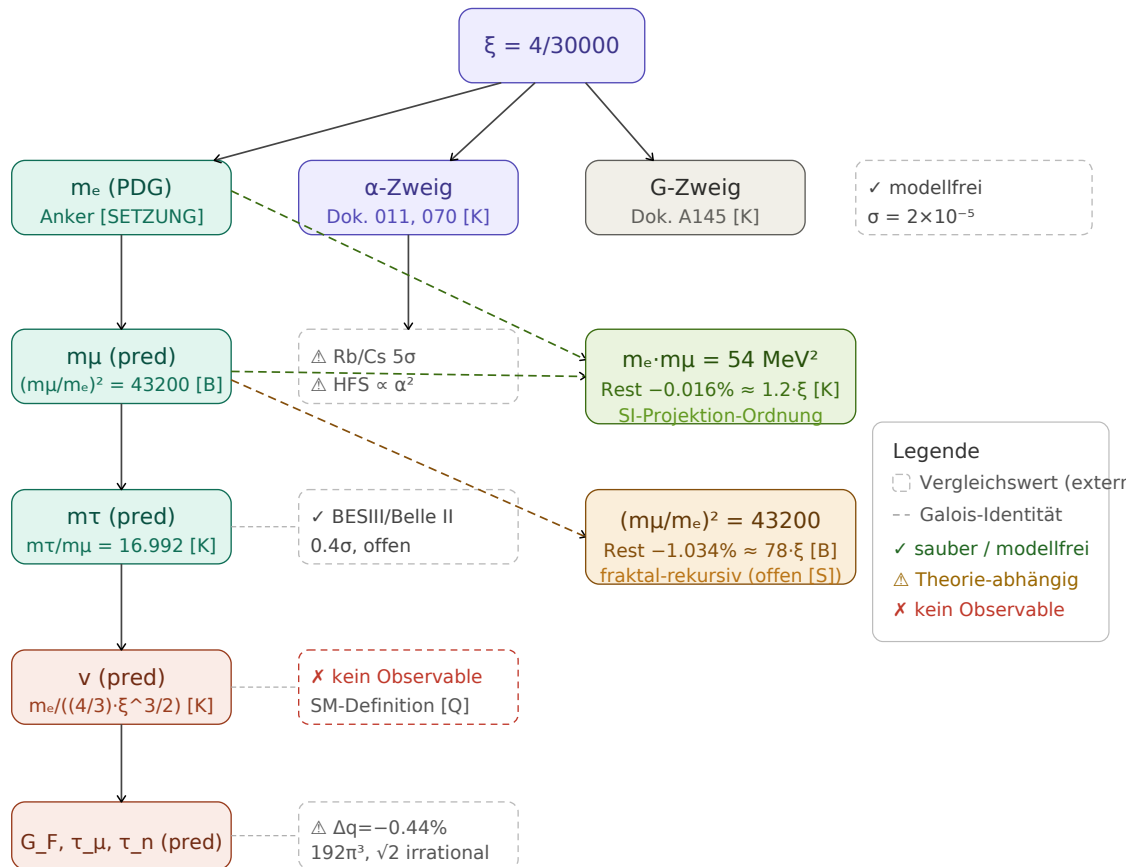
1. ν **is neither anchor nor observable.** Docs. 006, 319, R104: " $\nu = 246$ GeV (measured)" $\rightarrow$ " $\nu = 246$ GeV (SM definition from G_F)". Register R112.
2. m_μ/m_e **is QED-dependent.** "model-independent" $\rightarrow$ "least model-dependent available value"; $\nu_F \propto \alpha^2$. Register R113.
3. α **has no unique measured value below 1 ppb.** Cite the comparison value as CODATA average with Rb/Cs tension.
4. m_τ : two methods, PDG average 1776.93(9), FFGFT 0.4σ — open.
5. **New open item [S]:** τ_μ directly from FFGFT, without detour via G_F and ν — the only convention-free test of the weak sector.

Sources

All sources verified on 5 September 2026 against INSPIRE, APS, Nature, arXiv, or PubMed. Numerical values computed in exact rational arithmetic against CODATA 2022.

Bibliography

- [1] M. Eberhart, M. Fael, A. Keshavarzi, D. Nomura, K. Schönwald, M. Steinhauser, T. Teubner, J. Wright: *Muon lifetime and Fermi constant: an update*, arXiv:2607.02657 (2026).
- [2] V. Tishchenko et al. (MuLan): Phys. Rev. D **87**, 052003 (2013).
- [3] A. Sirlin: Phys. Rev. D **22**, 971 (1980).
- [4] M. I. Eides: *Muonium Hyperfine Splitting Uncertainty Revisited*, arXiv:2510.07281v2 (2026).
- [5] W. Liu et al.: Phys. Rev. Lett. **82**, 711 (1999).
- [6] M. I. Eides, H. Grotch, V. A. Shelyuto: Phys. Rep. **342**, 63 (2001).
- [7] L. Morel, Z. Yao, P. Cladé, S. Guellati-Khélifa: Nature **588**, 61 (2020).
- [8] R. H. Parker, C. Yu, W. Zhong, B. Estey, H. Müller: Science **360**, 191 (2018).
- [9] X. Fan, T. G. Myers, B. A. D. Sukra, G. Gabrielse: Phys. Rev. Lett. **130**, 071801 (2023).
- [10] T. Aoyama, T. Kinoshita, M. Nio: Atoms **7**, 28 (2019).
- [11] M. Ablikim et al. (BESIII): Phys. Rev. D **90**, 012001 (2014).
- [12] I. Adachi et al. (Belle II): Phys. Rev. D **108**, 032006 (2023).
- [13] V. V. Anashin et al. (KEDR): JETP Lett. **85**, 347 (2007).
- [14] PDG 2024, pdgLive S035M: τ mass average 1776.93 ± 0.09 MeV.
- [15] S. Sturm, F. Köhler, J. Zatorski et al.: Nature **506**, 467 (2014).
- [16] F. Köhler, S. Sturm, A. Kracke et al.: J. Phys. B **48**, 144032 (2015).
- [17] F. Heiße et al.: Phys. Rev. Lett. **131**, 253002 (2023).
- [18] Q. Li, C. Xue, J.-P. Liu et al.: Nature **560**, 582 (2018).
- [19] G. Rosi, F. Sorrentino, L. Cacciapuoti, M. Prevedelli, G. M. Tino: Nature **510**, 518 (2014).
- [20] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, T. Schwetz: JHEP **12** (2024) 216.
- [21] P. J. Mohr, D. B. Newell, B. N. Taylor, E. Tiesinga: Rev. Mod. Phys. **97**, 025002 (2025).



Derivation chains: from ξ to comparison values. Dashed boxes = external comparison values; dashed arrows = Galois identities.

1 Affected derivation chains

Which FFGFT chain depends on which comparison value, and what each finding changes.

Finding	Affected ments	docu- ments	Numerical rection	cor-	Wording correction
ν is SM definition (R112)	Docs. 006, R104	319,	none		" $\nu = 246$ GeV (measured)" $\rightarrow$ "(SM definition from G_F)"
$G_F: \Delta q, 192\pi^3, \sqrt{2}$	Docs. 006, 319	R104,	none		state $\Delta q = -0.44\%$; clarify with/without Δq
m_μ/m_e QED-dependent (R113)	Docs. 338, 011	317, 011	none		"model-independent" $\rightarrow$ "least model-dependent"
α Rb/Cs 5σ	Docs. 011, 338	070,	none		cite CODATA average with tension
m_τ two methods	Docs. 317, 006		none		cite BESIII + Belle II; PDG average 1776.93(9)
Neutrinos Δm^2	Docs. 339, 340		none		freeze NuFIT configuration
G	Doc. A145		none		—
Total			none		wording clarifications

The residual $(m_\mu/m_e)^2 = 43200$ vs. 42753.12 (difference 446.9) is $240\,000\sigma$ (CODATA) or $45\,000\sigma$ (real uncertainty, Eides) of the measurement uncertainty — physical (R113). No numerical value in the corpus changes.